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Question 1. Keeping Up With the Cartesians

(a)

$$B \times A = \{(1, 1), (2, 1)\}$$

$$C \times B = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2)\}$$

The claim holds.

(b)

$$B \times A = \{(1, 1), (1, 2)\}$$

$$C \times B = \{(1, 1), (2, 1), (3, 1)\}$$

The claim does not hold. $\{1, 2\}$ is an element of $B \times A$ but is not an element of $C \times B$.

(c)

Let (b, a) be an arbitrary tuple of (integer, integer).

Suppose that $(b, a) \in (B \times A)$. By the definition of Cartesian product, $b \in B$ and $a \in A$.

We are given that $A \subset B$. Since $a \in A$, by the definition of subset, $a \in B$.

We are also given that $B \subset C$. Since $b \in B$, by the definition of subset, $b \in C$.

Since $a \in B$ and $b \in C$, by the definition of Cartesian product, $(b, a) \in (C \times B)$.

Since (b, a) was arbitrary, we have proven, by definition of subset, that $(B \times A) \subset (C \times B)$.

Question 2. Our Finest Power

(a)

$$\mathcal{P}(A \cap (B \cup C)) = \mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

$$\mathcal{P}(A \cap B) \cup \mathcal{P}(A \cap C) = \mathcal{P}(\{1\}) \cup \mathcal{P}(\{2\}) = \{\emptyset, \{1\}\} \cup \{\emptyset, \{2\}\} = \{\emptyset, \{1\}, \{2\}\}$$

The claim does not hold. $\{1, 2\}$ is an element of $\mathcal{P}(A \cap (B \cup C))$ but is not an element of $\mathcal{P}(A \cap B) \cup \mathcal{P}(A \cap C)$.

(b)

$$\mathcal{P}(A \cap (B \cup C)) = \mathcal{P}(\{1\}) = \{\emptyset, \{1\}\}$$

$$\mathcal{P}(A \cap B) \cup \mathcal{P}(A \cap C) = \mathcal{P}(\{1\}) \cup \mathcal{P}(\{1\}) = \{\emptyset, \{1\}\} \cup \{\emptyset, \{1\}\} = \{\emptyset, \{1\}\}$$

The claim holds.

(c)

Let X be an arbitrary set.

Suppose $X \in \mathcal{P}(A \cap (B \cup C))$. By applying the Distributivity of Set Theorem, we get $X \in \mathcal{P}((A \cap B) \cup (A \cap C))$.

By the definition of power set, $X \subset ((A \cap B) \cup (A \cap C))$.

Let x be an arbitrary element of X . Since $X \subset ((A \cap B) \cup (A \cap C))$, then $x \in ((A \cap B) \cup (A \cap C))$.

By the definition of union, $x \in (A \cap B) \vee x \in (A \cap C)$. We continue by cases:

Suppose $x \in (A \cap B)$. Since we are given $(A \cap B) \subset (A \cap C)$, by the definition of subset, $x \in (A \cap C)$.

Suppose $x \in (A \cap C)$. Then $x \in (A \cap C)$ is already true.

Since our cases were exhaustive, $x \in (A \cap C)$. Then, since x was an arbitrary element of X , by the definition of subset, $X \subset (A \cap C)$.

By the definition of power set, since $X \subset (A \cap C)$, then $X \in \mathcal{P}(A \cap C)$.

By the definition of union, since $X \in \mathcal{P}(A \cap C)$, then $X \in \mathcal{P}(A \cap B) \cup \mathcal{P}(A \cap C)$.

Since X was arbitrary, we have proven, by the definition of subset, that $\mathcal{P}(A \cap (B \cup C)) \subset \mathcal{P}(A \cap B) \cup \mathcal{P}(A \cap C)$.

Question 3. Parmesan, Romano, and Meta

(a)

The claim is true.

Let x be an arbitrary element.

The stated biconditional holds since

$$\begin{aligned}
 x \in ((A \cap B) \cup (A \cap \overline{B})) &\equiv x \in (A \cap B) \vee x \in (A \cap \overline{B}) && \text{Def of } \cup \\
 &\equiv (x \in A \wedge x \in B) \vee (x \in A \wedge x \in \overline{B}) && \text{Def of } \cap \\
 &\equiv x \in A \wedge (x \in B \vee x \in \overline{B}) && \text{Distributivity} \\
 &\equiv x \in A \wedge T && \text{Excluded Middle} \\
 &\equiv x \in A && \text{Identity}
 \end{aligned}$$

Since x was arbitrary, we have proven, by the definition of set equality, that $(A \cap B) \cup (A \cap \overline{B}) = A$.

(b)

The claim is false.

Suppose $A = \emptyset$, $B = \{1\}$, and $C = \emptyset$.

$$(A \cup B) \setminus C = \{1\} \setminus \emptyset = \{1\}$$

$$(A \cup C) \setminus B = \emptyset \setminus \{1\} = \emptyset$$

Since a counterexample exists, the claim is false.

(c)

The claim is true.

Let x be an arbitrary element.

The stated biconditional holds since

$$\begin{aligned}
 x \in (A \setminus (B \cap C)) &\equiv x \in A \wedge \neg(x \in (B \cap C)) && \text{Def of set difference} \\
 &\equiv x \in A \wedge \neg(x \in B \wedge x \in C) && \text{Def of } \cap \\
 &\equiv x \in A \wedge (x \notin B \vee x \notin C) && \text{DeMorgan's Law} \\
 &\equiv (x \in A \wedge x \notin B) \vee (x \in A \wedge x \notin C) && \text{Distributivity} \\
 &\equiv x \in (A \setminus B) \vee x \in (A \setminus C) && \text{Def of set difference} \\
 &\equiv x \in ((A \setminus B) \cup (A \setminus C)) && \text{Def of } \cup
 \end{aligned}$$

Since x was arbitrary, we have proven, by the definition of set equality, that $(A \setminus (B \cap C)) = (A \setminus B) \cup (A \setminus C)$.

Question 4. Optional Practice Problems

(a)

i. Let x be an arbitrary element of A .

Then by the definition of subset, $\{x\} \subset A$.

Then, by the definition of powerset, $\{x\} \in \mathcal{P}(A)$.

Since we were given that $\mathcal{P}(A) \subset \mathcal{P}(B)$, then by the definition of subset, $\{x\} \in \mathcal{P}(B)$.

Then, by the definition of powerset, $\{x\} \subset B$.

Which means that by the definition of subset, $x \in B$.

Therefore, since x was arbitrary, we have proven, by the definition of subset, that $A \subset B$.

iii. Let (a, b) be an arbitrary element of $A \times B$.

Then, by the definition of Cartesian product, $a \in A$ and $b \in B$.

Since we are given $A \subset B$, then by the definition of subset, $a \in B$.

Since $a \in B$ and $b \in B$, then by the definition of Cartesian product, $(a, b) \in B \times B$.

Therefore, since (a, b) was arbitrary, we have proven, by the definition of subset, that $A \subset B$.

(b)

i. The claim is false.

Suppose $A = \{1\}$, $B = \{1\}$, and $C = \emptyset$.

$$A \setminus (B \cap C) = \{1\} \setminus \emptyset = \{1\}$$

$$(A \setminus B) \cap (A \setminus C) = \emptyset \cap \{1\} = \emptyset$$

Since a counterexample exists, the claim is false.

iii. Let x be an arbitrary element.

The stated biconditional holds since

$$\begin{aligned}
 x \in ((A \cap \overline{A}) \cup \overline{B}) \cup \overline{C} &\equiv x \in (A \cap \overline{A} \vee x \in \overline{B}) \vee x \in \overline{C} && \text{Def of } \cup \\
 &\equiv (x \in A \wedge x \notin A) \vee x \in \overline{B} \vee x \in \overline{C} && \text{Def of } \cap \\
 &\equiv F \vee x \in \overline{B} \vee x \in \overline{C} && \text{Contradiction} \\
 &\equiv x \in \overline{B} \vee x \in \overline{C} && \text{Identity} \\
 &\equiv x \in (\overline{B} \cup \overline{C}) && \text{Def of } \cup \\
 &\equiv x \in \overline{B \cap C} && \text{DeMorgan's Law}
 \end{aligned}$$

Therefore, since x was arbitrary, we have proven that $((A \cap \overline{A}) \cup \overline{B}) \cup \overline{C} = \overline{B \cap C}$ by the definition of set equality.

(c)

i. Claim: Let $P(L)$ be “ $\text{len}(\text{double}(L)) = 2 \cdot \text{len}(L)$ ”. We will show that $P(L)$ holds for all Lists L by induction on L .

Base Case: $\text{len}(\text{double}(\text{nil})) = \text{len}(\text{nil}) = 0 = 2 \cdot 0 = 2 \cdot \text{len}(\text{nil})$, so $P(\text{nil})$ is true.

Inductive Hypothesis: Suppose $P(L)$ is true for some arbitrary $L \in \text{List}$.

Inductive Step:

$$\begin{aligned} \text{len}(\text{double}(a :: L)) &= \text{len}(a :: a :: \text{double}(L)) && \text{Def of double} \\ &= 1 + \text{len}(a :: \text{double}(L)) && \text{Def of len} \\ &= 1 + 1 + \text{len}(\text{double}(L)) && \text{Def of len} \\ &= 2 + \text{len}(\text{double}(L)) && \text{Algebra} \\ &= 2 + 2 \cdot \text{len}(L) && \text{I.H.} \\ &= 2 \cdot (1 + \text{len}(L)) && \text{Algebra} \\ &= 2 \cdot (\text{len}(a :: L)) && \text{Def of len} \end{aligned}$$

Thus $P(a :: L)$ is true.

Conclusion: Therefore, we have shown that $P(L)$ is holds for all lists L by induction.